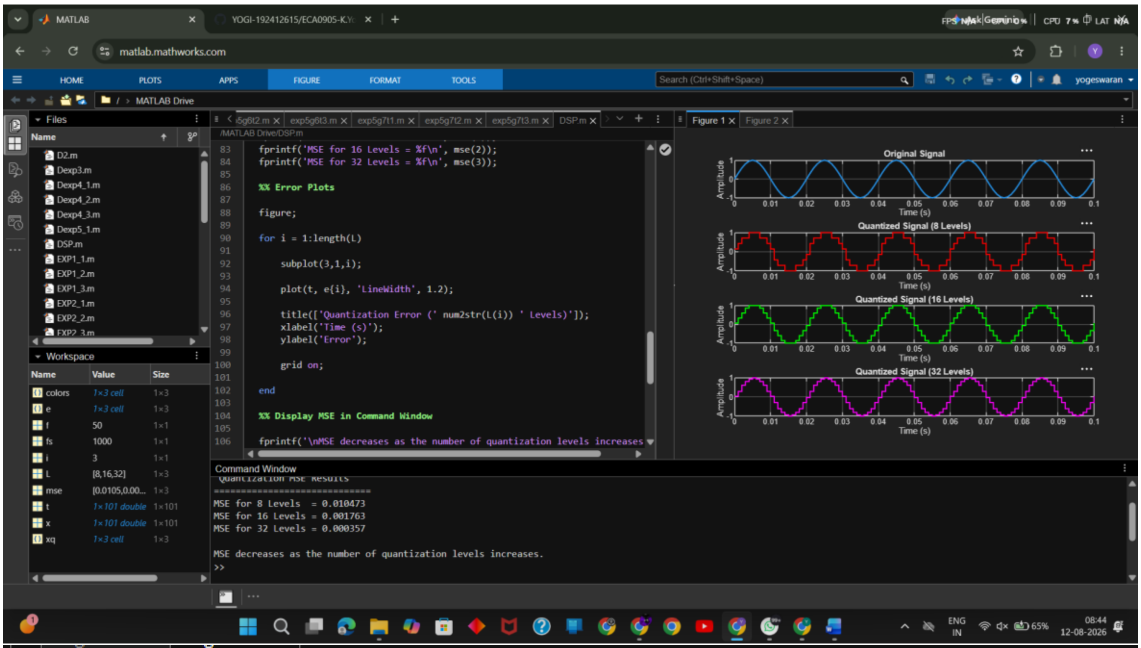
EXP 6 Finite Word length Effects
(6.1)Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

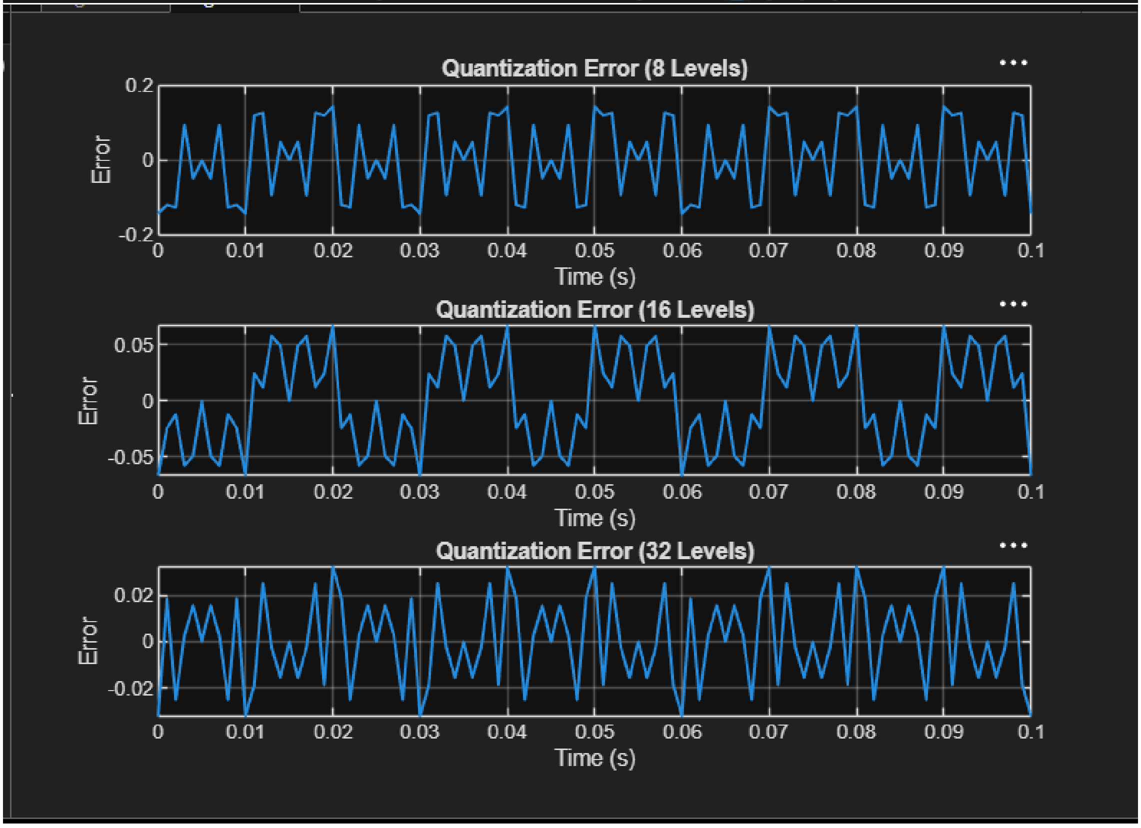
Program

```
clc;
clear;
close all;

%% Signal Parameters
fs = 1000;              % Sampling frequency
f = 50;                 % Signal frequency
t = 0:1/fs:0.1;        % Time vector

% Original sinusoidal signal
x = sin(2*pi*f*t);

%% Quantization Levels
L = [8 16 32];

% Colors for different quantization levels
colors = {'r', 'g', 'm'};

% Preallocate variables
xq = cell(1, length(L));
e = cell(1, length(L));
mse = zeros(1, length(L));

%% Quantization and MSE Calculation

for i = 1:length(L)

    % Quantization
    xq{i} = round((x + 1) * (L(i) - 1) / 2) ...
            * (2 / (L(i) - 1)) - 1;

    % Quantization error
    e{i} = x - xq{i};

    % Mean Square Error
    mse(i) = mean(e{i}.^2);

end

%% Plot Original and Quantized Signals

figure;

% Original Signal
subplot(4,1,1);
plot(t, x, 'LineWidth', 1.5);
title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Quantized Signal - 8 Levels
subplot(4,1,2);
stairs(t, xq{1}, colors{1}, 'LineWidth', 1.2);
title('Quantized Signal (8 Levels)');
```

```matlab
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Quantized Signal - 16 Levels
subplot(4,1,3);
stairs(t, xq{2}, colors{2}, 'LineWidth', 1.2);
title('Quantized Signal (16 Levels)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Quantized Signal - 32 Levels
subplot(4,1,4);
stairs(t, xq{3}, colors{3}, 'LineWidth', 1.2);
title('Quantized Signal (32 Levels)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

%% Display MSE Values

fprintf('\n============================\n');
fprintf(' Quantization MSE Results\n');
fprintf('============================\n');

fprintf('MSE for 8 Levels  = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));

%% Error Plots

figure;

for i = 1:length(L)

    subplot(3,1,i);

    plot(t, e{i}, 'LineWidth', 1.2);

    title(['Quantization Error (' num2str(L(i)) ' Levels)']);
    xlabel('Time (s)');
    ylabel('Error');

    grid on;

end

%% Display MSE in Command Window

fprintf('\nMSE decreases as the number of quantization levels increases.\n');
```



fprintf('MSE for 16 Levels = %f\n', mse(2));
fprintf('MSE for 32 Levels = %f\n', mse(3));
%% Error Plots
figure;
for i = 1:length(L)
subplot(3,1,i);
plot(t, e{i}, 'LineWidth', 1.2);
title(['Quantization Error (' num2str(L(i)) ' Levels)']);
xlabel('Time (s)');
ylabel('Error');
grid on;
end
%% Display MSE in Command Window
fprintf('\nMSE decreases as the number of quantization levels increases
Command Window
MSE for 8 Levels = 0.010473
MSE for 16 Levels = 0.001763
MSE for 32 Levels = 0.000357
MSE decreases as the number of quantization levels increases.
Original Signal
Quantized Signal (8 Levels)
Quantized Signal (16 Levels)
Quantized Signal (32 Levels)



Quantization Error (8 Levels)
Error
Time (s)
Quantization Error (16 Levels)
Error
Time (s)
Quantization Error (32 Levels)
Error
Time (s)

(6.2)Analysis of Finite Word Length Effects in Digital Signal Representation Using Truncation and Rounding Techniques in MATLAB

Program

```
clc;
clear;
close all;

%% Signal Generation
n = 0:79;

% Original sinusoidal signal
x = sin(2*pi*n/40);

% Quantization factor
q = 8;

%% Quantization

% Truncation quantization
xt = floor(x*q)/q;

% Rounding quantization
xr = round(x*q)/q;

%% Quantization Errors

% Error due to truncation
et = x - xt;

% Error due to rounding
er = x - xr;

%% Fig 1: Original vs Quantized Signal

figure;

plot(n, x, 'b', 'LineWidth', 1.5);
hold on;
plot(n, xr, 'r', 'LineWidth', 1.5);

grid on;

title('Original vs Quantized Signal');
xlabel('Sample Number');
ylabel('Amplitude');

legend('Original', 'Quantized');

%% Fig 2: Truncation vs Rounding

figure;

plot(n, x, 'b', 'LineWidth', 1.5);
hold on;

plot(n, xt, 'r', 'LineWidth', 1.5);
plot(n, xr, 'g', 'LineWidth', 1.5);

grid on;
```

```
title('Truncation vs Rounding');
xlabel('Sample Number');
ylabel('Amplitude');

legend('Original', 'Truncated', 'Rounded');

%% Fig 3: Quantization Error

figure;

stem(n, et, 'r', 'filled');
hold on;

stem(n, er, 'g', 'filled');

grid on;

title('Quantization Error');
xlabel('Sample Number');
ylabel('Error');

legend('Truncation', 'Rounding');

%% Fig 4: MSE Comparison

% Calculate Mean Square Error
mse_truncation = mean(et.^2);
mse_rounding = mean(er.^2);

figure;

bar([mse_truncation mse_rounding]);

set(gca, 'XTick', 1:2);
set(gca, 'XTickLabel', {'Truncation', 'Rounding'});

grid on;

title('MSE Comparison');
ylabel('MSE');

%% Display MSE Values

fprintf('\n============================\n');
fprintf(' Quantization MSE Results\n');
fprintf('============================\n');

fprintf('MSE using Truncation = %f\n', mse_truncation);
fprintf('MSE using Rounding   = %f\n', mse_rounding);

fprintf('\nLower MSE indicates better quantization accuracy.\n');
```

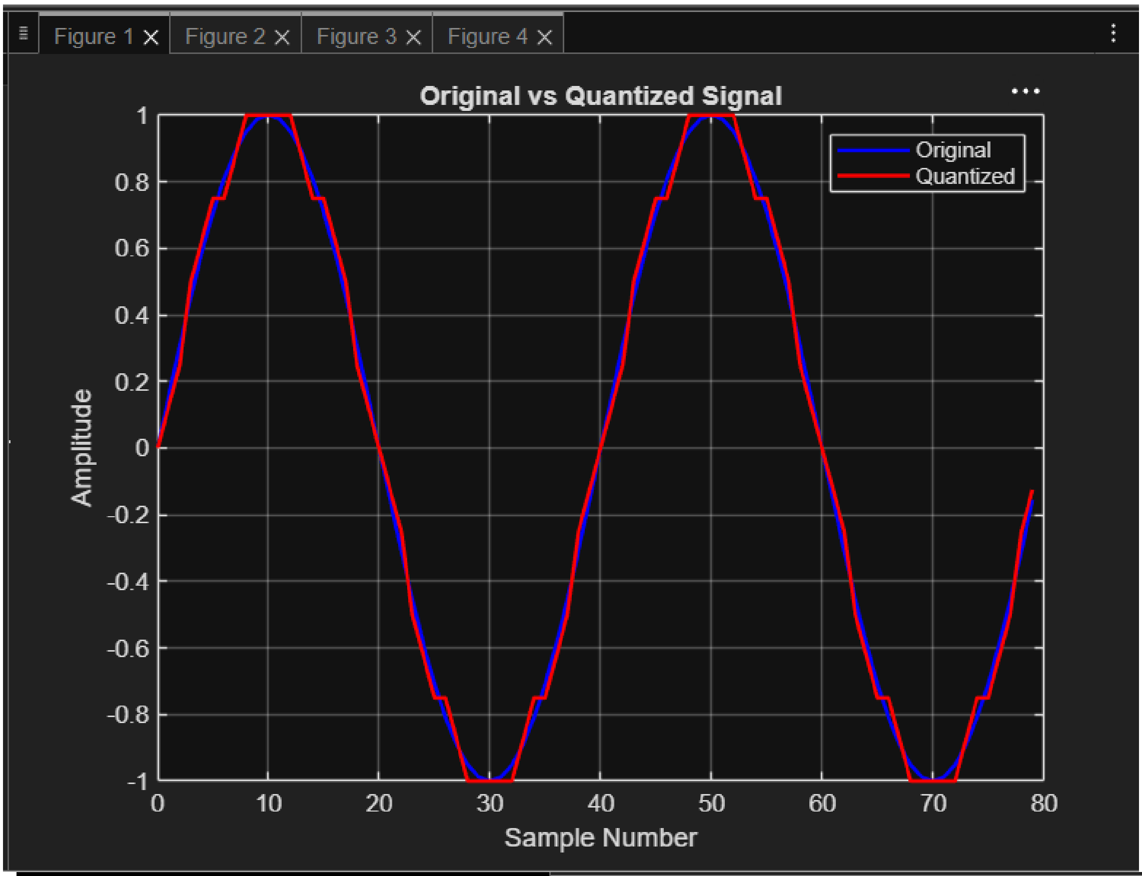


Figure 1
Figure 2
Figure 3
Figure 4
Original vs Quantized Signal
Original
Quantized
Amplitude
Sample Number
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
10
20
30
40
50
60
70
80

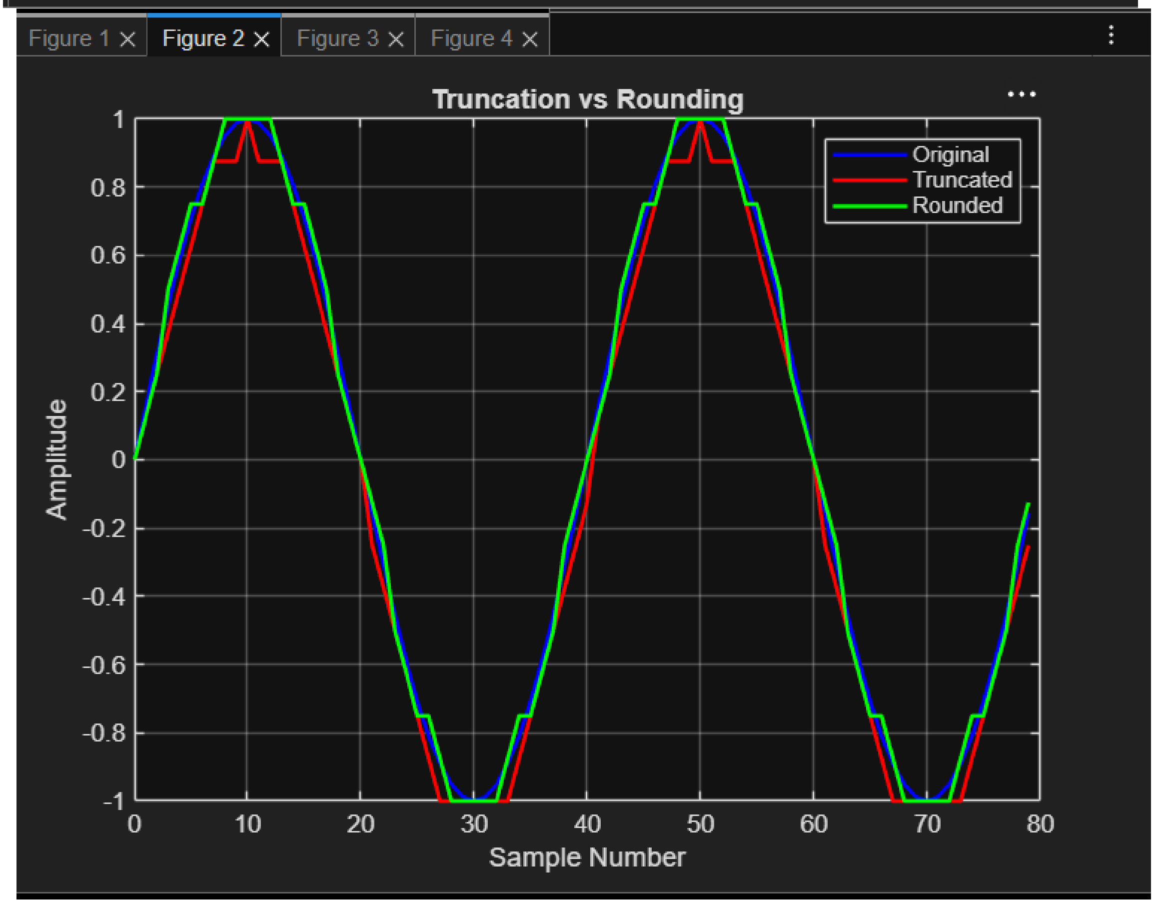


Figure 1
Figure 2
Figure 3
Figure 4
Truncation vs Rounding
Original
Truncated
Rounded
Amplitude
Sample Number
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
10
20
30
40
50
60
70
80

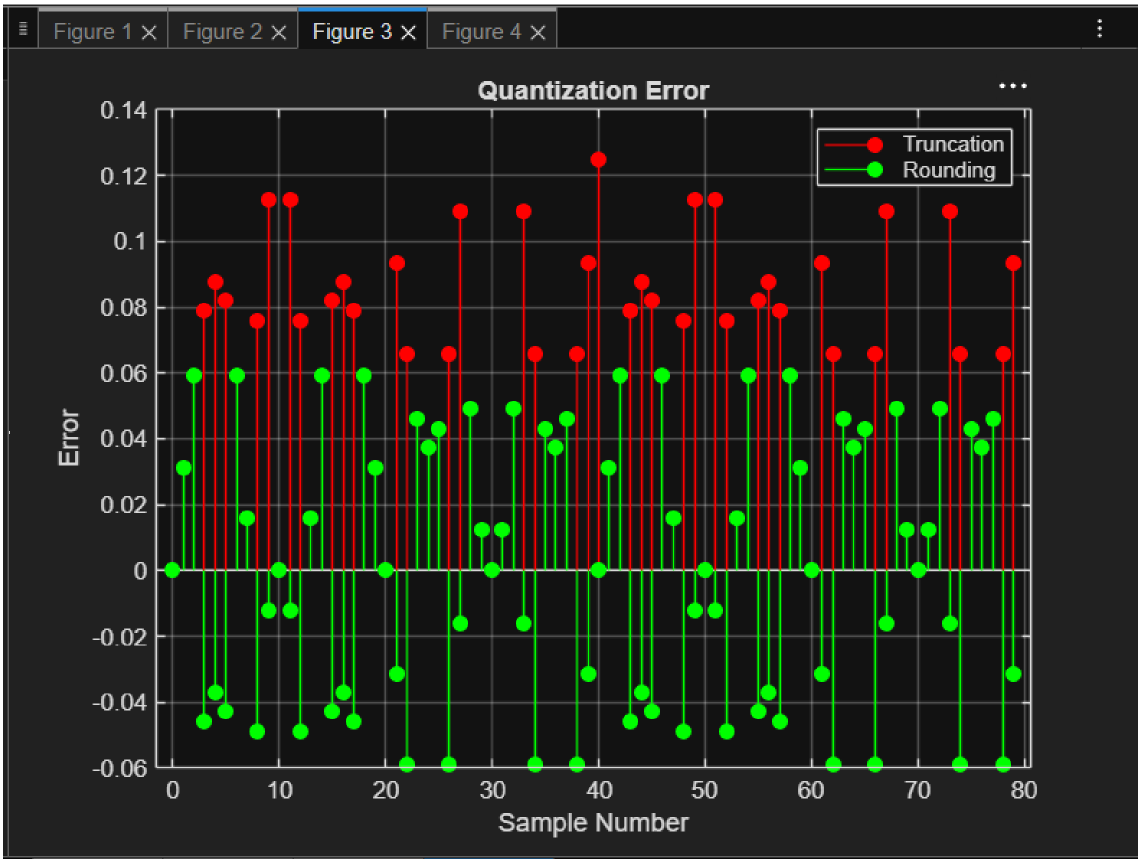


Figure 1
Figure 2
Figure 3
Figure 4
Quantization Error
Truncation
Rounding
Error
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
-0.02
-0.04
-0.06
0
10
20
30
40
50
60
70
80
Sample Number

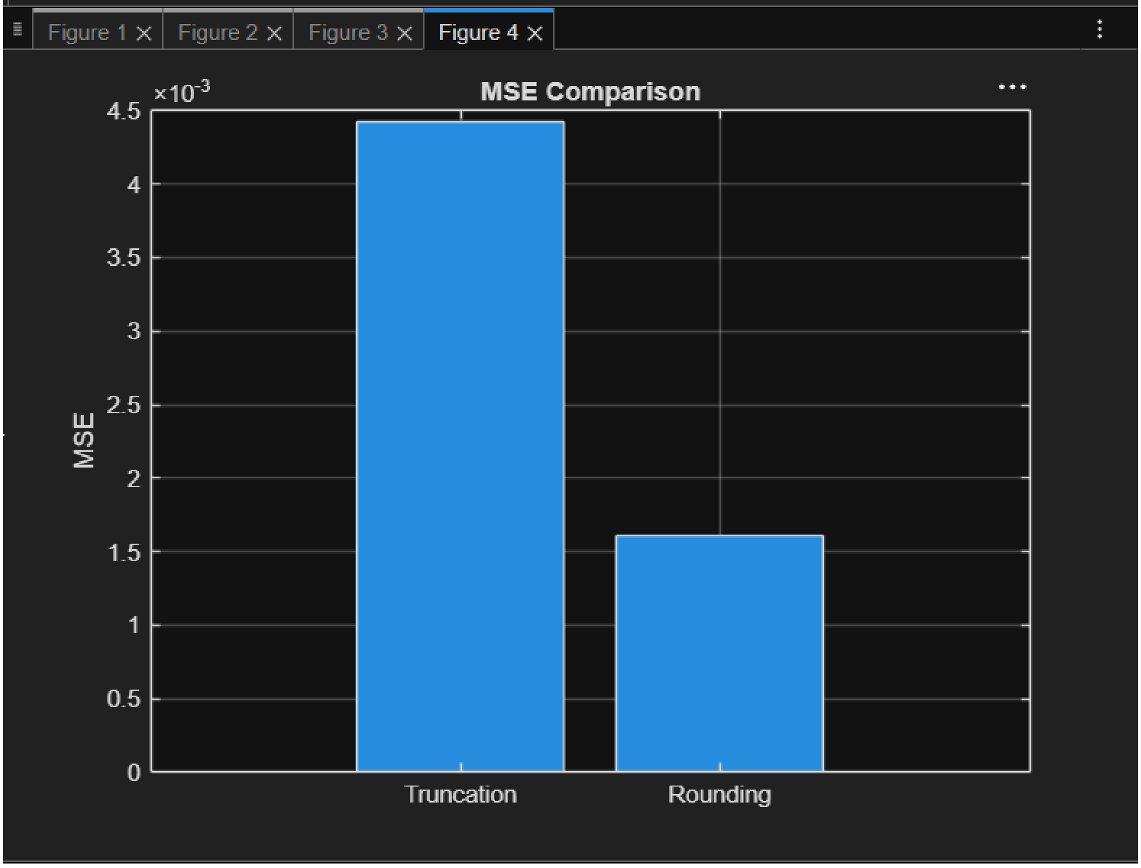


Figure 1
Figure 2
Figure 3
Figure 4
MSE Comparison
×10^-3
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
MSE
Truncation
Rounding